NIVRA

Before impact.

Bal Bharati Hackathon 3.0 / Idea Submission

PROBLEM STATEMENT

Developing incidents can go unnoticed; camera footage alone cannot bring help.

Theme: T3 — AI for Good

Team: Damage Control

School: Bal Bharati Public School, Pitampura

Team leader: _____

Team size (excluding mentor): 5 Category: ____



Your Idea

AI-assisted early intervention, with a human in charge.





Idea / Solution

Use a phone or camera feed to flag visible movement patterns that may need attention.

Problem Resolution

Give staff a short replay and location, then connect a confirmed alert to a responder.

Observe → flag → human review

Decide → dispatch → acknowledge

Warning patterns, seen over time

YOUR IDEA / WHAT NIVRA WOULD LOOK FOR







Closing distance

People rapidly move closer.

Aggressive gestures

Sudden, repeated body movements.

Crowd convergence

Several people gather around an event.

Multiple cues would request a review. A movement pattern alone does not establish intent.

Eight seconds of context. A human decision.

YOUR IDEA / PROPOSED OPERATOR WORKFLOW



Review the previous 8 seconds

See the location and warning signals, then choose the next action.

Dismiss

Close a false or harmless alert.

Monitor

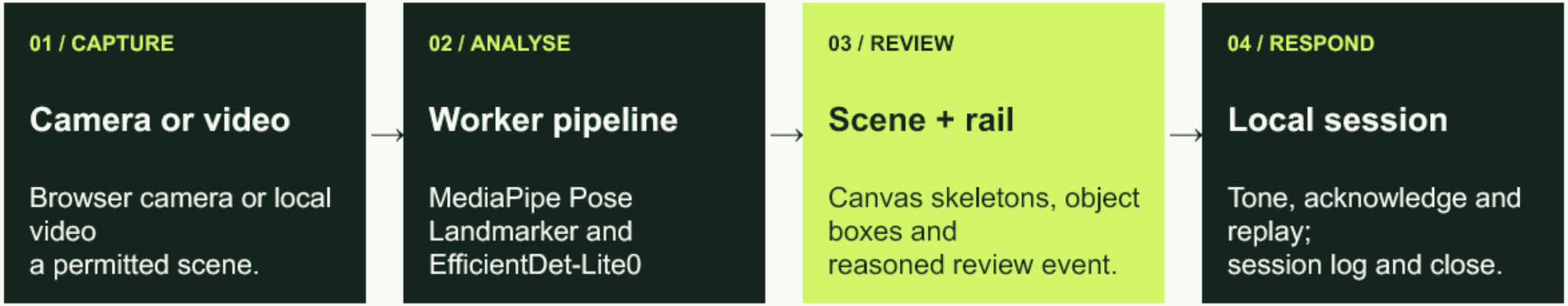
Keep watching when context is unclear.

Dispatch

Send an available responder and record acknowledgement.

Technical Approach

Proposed architecture / one supervised demo



Frontend / runtime

React 19 + TypeScript + Vite. Canvas draws aligned skeletons and object boxes.

Workers / models

Two MediaPipe workers isolate pose runtimes; an EfficientDet-Lite0 CPU worker runs object analysis. No backend or cloud service is required.

Local browser inference. Frames, tracks and review events stay in the session.

Feasibility and Viability

Technical

MediaPipe and EfficientDet assets are bundled locally. Pose and object rules are implemented; real-world accuracy remains unvalidated.

Operational

Camera, local video or scripted test scenario. Operator reviews one event at a time and acknowledges it.

Resources needed

Node 24, Chrome, browser camera/video, local model assets and a capable CPU/GPU. No cloud backend.

Data & build plan

Frames are not uploaded, recorded or persisted. Use permitted staged footage and the repository's scripted scenarios.

Challenge / risk	Proposed mitigation
Ordinary play creates false alerts	Use temporal rules; require human review before dispatch.
Occlusion, poor light or network loss	Show confidence / offline state; test varied scenes.
Video privacy and child data	Minimise retention and access; obtain school approval and review applicable rules.

Build small. Test the complete response.

FEASIBILITY / PROPOSED VALIDATION PLAN

01 **Connect**

Open camera or local video in Chrome; no cloud connection.

02 **Stage**

Use permitted footage or the repository's scripted test scenes.

03 **Review**

Test alerts, the 8-second replay and operator choices.

04 **Respond**

Deliver a confirmed alert; record acknowledgement and closure.

Measure 211 automated tests • build output • false alerts • missed staged events • review time

Impact and Benefits

Why earlier attention matters.

The supplied public violence metrics report frames a large human and economic burden.

1 in 6

high-school students report a physical fight within 12 months.

12,000 / day

people reportedly lose their lives worldwide to trauma and violence.

$20.3T–$30T

estimated global annual cost of interpersonal violence.

NIVRA's intended contribution

Move from passive recording to observable signals, human review and coordinated response.

Guardrails

Local processing • no identity inference • human decision before action

Source: supplied public_violence_metrics_report.pdf. Figures are reproduced as reported and were not independently verified.

Research and References

Evidence informs the idea; it does not validate NIVRA.

01 / SCHOOL SAFETY

MoE, School Safety & Security Guidelines (2021).
Establishes institutional safety responsibilities.

02 / TECHNICAL BASIS

Google Research, BlazePose (2020), and MediaPipe docs.
Support pose estimation—not prediction of violence.

03 / EXISTING SOLUTION

Axis Object Analytics: object tracking and event triggers.
NIVRA proposes human review plus response acknowledgement.

04 / DATA & PRIVACY

Planned data: permissioned staged volunteer videos; none yet.
DPDP Act (2023): assess applicable privacy rules before a pilot.

Proposed stack

React + TypeScript + Vite

React 19 UI, TypeScript 6 and Vite 8 build.

MediaPipe Pose Landmarker

Full/Lite float16 bundles via @mediapipe/tasks-vision.

EfficientDet-Lite0

Lower-frequency CPU worker for local object analysis.

Web Workers + WASM

Separate pose workers, GPU attempt and CPU fallback; no backend.

Team Member Details

Bal Bharati Public School, Pitampura

Students

SL	Student ID	Name	Class	DOB	Email	Phone Number	Gender
1	223564	Harsh Jain	12C	23/04/09	223564@pp.balbharati.org	9810141505	Male
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Mentor

SL	Name	UID	Address	DOB	Email	Phone Number	Gender
1							